

Graph Ratio Tables

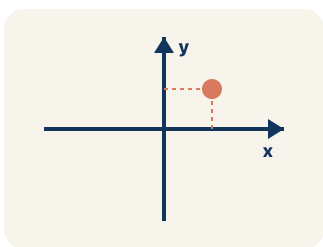
Lesson 3-3

Name: _____ Date: _____ Class: _____

Key Vocabulary

Level 1 support

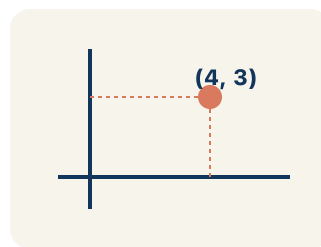
Picture first, then the word, then a plain-language meaning. Say each word out loud.



A + shape made by two number lines crossing at the center point $(0, 0)$

Coordinate plane

A grid with a line going across and a line going up to plot points.



$(4, 3)$ means go right 4, up 3

Ordered pair

Two numbers (x, y) that tell where a point is on a grid.



$(1,2), (2,4), (3,6)$ all line up

Linear pattern

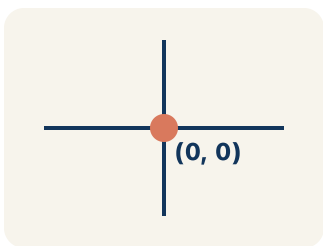
Points that make a straight line on a grid.

$$\frac{1}{2} = \frac{2}{4}$$

A straight line from $(0,0)$ through $(2,6)$ and $(4,12)$

Proportional

Two amounts that grow together at the same rate.



The starting corner of the grid at $(0, 0)$

Origin

The point $(0, 0)$ where the two grid lines cross.

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can graph the values from a ratio table as points on the coordinate plane.

1 Notice

Chef Academy students are tracking how much chocolate sauce they need for different numbers of sundaes.

2 Key idea

I can graph the values from a ratio table as points on the coordinate plane.

3 Words I'll use

Coordinate plane

Ordered pair

Linear pattern

Proportional

Origin



Think About It

- What two quantities could we put on the x-axis and y-axis?
- What pattern do you see in the ratio table values?
- What do you think the graph will look like?

See the notes in action: watch one worked all the way through, then try the next with the same steps.



I do — watch

Follow each step as your teacher solves it.

Problem: A ratio table shows (2, 6), (4, 12), and (6, 18). Which ordered pair comes next if the pattern continues?

- A. (8, 24)
- B. (7, 20)
- C. (8, 22)
- D. (10, 24)

Step 1

The pattern adds 2 to x and 6 to y each time.

Step 2

After (6, 18): $x = 6 + 2 = 8$, $y = 18 + 6 = 24$.

Step 3

The next point is (8, 24).



Answer: A. (8, 24)

 We do — together

Solve this one with your class using the same steps.

Problem: When you graph equivalent ratios, the points will always form what shape?

- A. A straight line through the origin
- B. A curved line
- C. A zigzag
- D. A circle

Step 1

Step 2

Step 3

Answer:

 You do — your turn

Now try one on your own. Show every step.

Problem: A ratio table shows (2, 8), (3, 12), and (5, 20). What is the y-value when $x = 7$?

- A. 28
- B. 24
- C. 21
- D. 35

Show your work:

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

The sundae recipe uses 2 ounces of chocolate sauce for every 1 sundae. Before you graph, what do you predict the points will look like, and which quantity goes on each axis?

Level 1 support

Start here: I predict the points will line up because sundaes and sauce grow together.

Need a hint?

Sentence starters (optional)

I will put ____ on the x-axis and ____ on the y-axis.

Pondré ____ en el eje x y ____ en el eje y.

I predict the points will form a ____.

Predigo que los puntos formarán una ____.

Word bank: Coordinate plane Ordered pair Origin

Listen for: A strong answer chooses sundaes (x) and ounces of sauce (y), and predicts a straight line because each sundae always needs 2 more ounces.

Level 2

Predict whether the line will pass through the origin (0, 0). Justify your prediction using what 0 sundaes would mean.

- The line will pass through the origin because ____ sundaes means ____ ounces.
- I can justify this because the ratio stays ____ even at zero.

EXPLORE

After plotting (1,2), (2,4), (3,6)... what pattern do you notice, and how does it show the chocolate-sauce ratio?

Level 1 support

Start here: The points form a straight line, and each sundae adds 2 ounces of sauce.

 **Need a hint?**

Sentence starters (optional)

The points form a ____ through the ____.

Los puntos forman una ____ que pasa por el ____.

As the number of sundaes goes up by 1, the ounces go up by ____.

Cuando el número de copas sube en 1, las onzas suben en ____.

Word bank: Coordinate plane Ordered pair Linear pattern Proportional Origin

Listen for: Listen for 'straight line through the origin' and the connection that the constant 2-ounce step shows a proportional relationship.

Level 2

Critique this claim: 'Any points that go up to the right form a proportional graph.'

What would the points (1,2),(2,4),(3,7) show instead, and why?

- The claim is wrong because a proportional graph must also ____.
- The points (1,2),(2,4),(3,7) are NOT proportional because ____.

CONNECT

The food truck plots tacos vs. salsa and gets a straight line. What does that graph tell the owner about preparing salsa for ANY number of tacos?

Level 1 support

Start here: The straight line lets the owner predict salsa for any taco amount.

 Need a hint?

Sentence starters (optional)

This is like our graphing work because ____ and ____ are related by ____.

Esto es como nuestro trabajo con gráficas porque ____ y ____ se relacionan por ____.

The graph lets the owner predict ____.

La gráfica permite al dueño predecir ____.

Word bank: Coordinate plane Ordered pair Proportional Origin

Listen for: A strong answer says the proportional line lets the owner read off (or extend to) salsa amounts for taco counts not in the table.

Level 2

Generalize: if the food-truck line were steeper than another truck's line, what would that tell you about how much salsa each truck uses per taco?

- A steeper line means ____ salsa per taco because ____.
- I can compare the two trucks by looking at ____.

Write About the Math

The Writing Revolution

I can explain my graph using the words coordinate plane, ordered pair, origin, and proportional.

1. Kernel Sentence subject + verb

Model: Coordinate plane is a grid with a line going across and a line going up to plot points.

Plano cartesiano es una cuadrícula con una línea horizontal y una vertical para marcar puntos.

Write a kernel sentence about coordinate plane. Use a subject and a verb.

Escribe una oración base sobre plano cartesiano. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Coordinate plane matters in math

Plano cartesiano importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Coordinate plane matters in math because ____.

Plano cartesiano importa en matemáticas porque ____.

but
pero

Coordinate plane matters in math, but ____.

Plano cartesiano importa en matemáticas, pero ____.

so
entonces

Coordinate plane matters in math, so ____.

Plano cartesiano importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about coordinate plane.
Di un hecho verdadero sobre coordinate plane.

Coordinate plane ____.

Question *Pregunta*

Ask a question about coordinate plane.
Haz una pregunta sobre coordinate plane.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about coordinate plane.
Muestra entusiasmo sobre coordinate plane.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with coordinate plane.
Dile a un compañero qué hacer con coordinate plane.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

I will put ____ **on the x-axis** and ____ **on the y-axis.**

Pondré ____ *en el eje x* y ____ *en el eje y.*

I predict the points will form a ____.

Predigo que los puntos formarán una ____.

The points form a ____ **through the** ____.

Los puntos forman una ____ *que pasa por el* ____.

Try It

Solve on your own. Check the answer key when you are done.

1. A line through the origin contains (3, 9). What is the unit rate (y per 1 x)?

- A. 3
- B. $\frac{1}{3}$
- C. 6
- D. 9

Show your work:

2. Why do graphs of equivalent ratios start at the origin (0,0)?

- A. Because 0 of one quantity pairs with 0 of the other
- B. Because every line starts at zero
- C. Because ratios are always negative
- D. Because the y-axis is longer

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A student plots the points (1, 3), (2, 6), (3, 9), and (4, 15) from a ratio table. She says they form a straight line because they go up. Is she correct? Explain how to check.

Show your work:

Reflect — Exit Ticket

A ratio table shows cups of rice to cups of water: (1, 2), (2, 4), (3, 6). If you plot these points, the line passes through which point?

- A. (5, 10)
- B. (4, 6)
- C. (5, 8)
- D. (6, 10)

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. A straight line through the origin — *Equivalent ratios are proportional, so their graph is always a straight line that passes through the origin (0, 0).*
2. **You Try (notes):** A. 28 — *The ratio is $y/x = 8/2 = 4$. For every x , $y = 4x$. When $x = 7$: $y = 4 \times 7 = 28$.*
3. **Try It 1:** A. 3 — *$9 \div 3 = 3$, so y increases by 3 for each 1 unit of x .*
4. **Try It 2:** A. Because 0 of one quantity pairs with 0 of the other — *When you have 0 of one amount you have 0 of the other, so the line passes through (0,0).*
5. **Exit Ticket:** A. (5, 10) — *The ratio is 1:2, so for 5 cups of rice you need $5 \times 2 = 10$ cups of water. The point (5, 10) continues the pattern.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Coordinate plane is a grid with a line going across and a line going up to plot points.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).